PRISM Energy Intelligence

NTPC generation and investment research

A working prototype with dated evidence, historical evaluation, financial context and a grounded research assistant.

1.50%

day-60 generation MAPE

8 quarters

historical comparisons

53 reports

unique CEA originals

Developed by Sutanu Chatterjee (Trial Task for PRISM)

The project answers each part of the brief

Brief requirement	Completed project response
Define the problem	Help an analyst assess early operating momentum and the financial case for NTPC.
Build a working prototype	React dashboard, public reports, a fixed station sample, transparent calculations and dated inputs.
Present the result	Generation evaluation, financial assessment, valuation scenarios, source audits and Prismy.
Document the work	README, source manifests, tests, setup, findings, limitations and the next improvements.

The scope prioritises inspectable research and judgment over a production trading model.

Two evidence streams meet in the research view

Operating study

Public CEA reports
Same eight thermal station IDs

Historical cutoff
Only eligible captured reports

Quarter estimate
Run rate and two comparison baselines

Later outcome scoring
Eight quarters and all three cutoffs

Financial snapshot

Consolidated filings
Annual results and June-quarter results

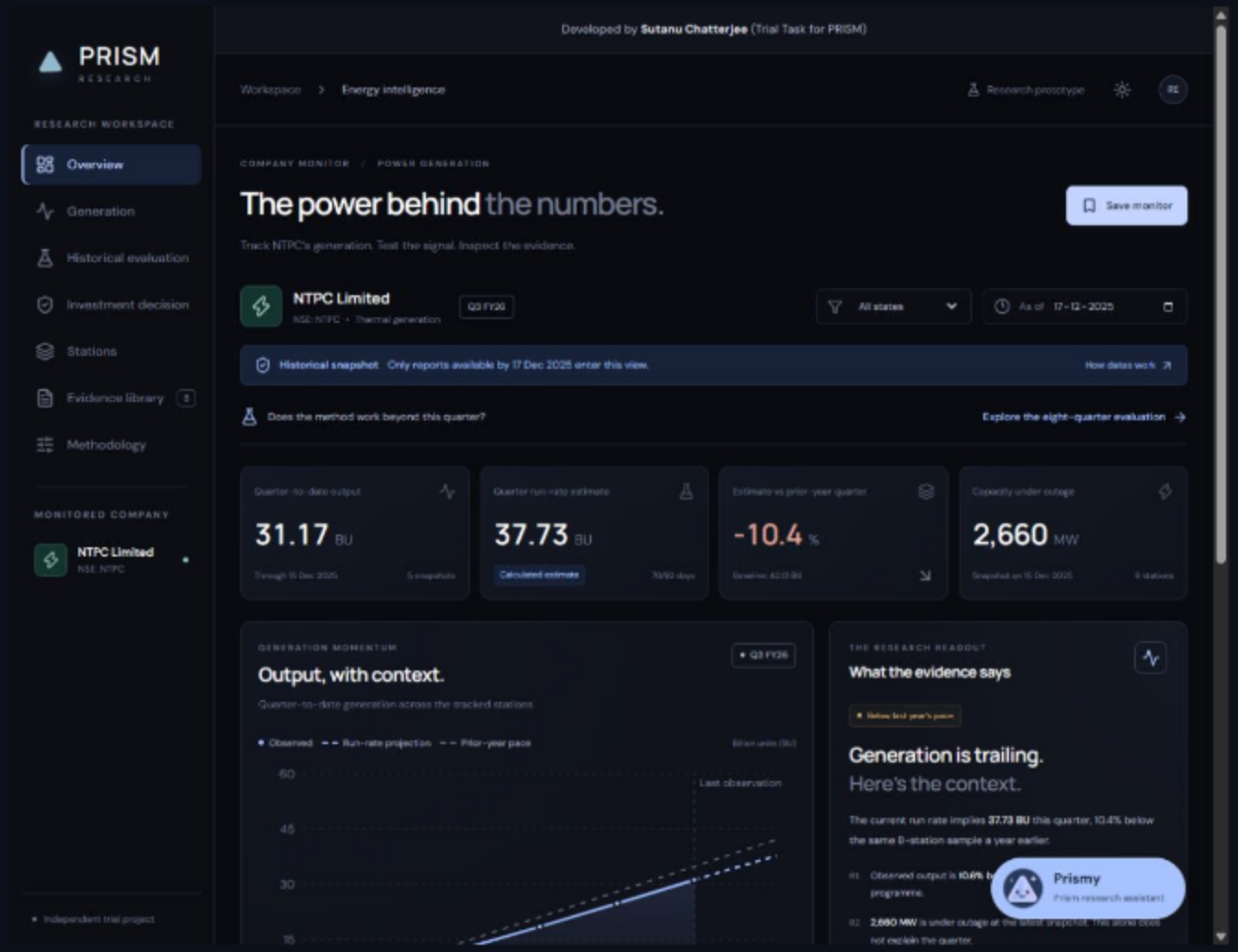
Earnings, cash and debt
Tax effects and financing interest

Dated NSE close
28 October 2026 price snapshot

Valuation assumptions
Earnings haircut, P/E and safety margin

Research assessment and Prismy’s source-linked answers

The dashboard makes evidence easy to inspect



Seven research views

Operations, evaluation, financials, sources and methodology.

Traceable outputs

Open preserved PDFs, inspect source rows and export results.

Simple handoff

Static hosted demo and reproducible local setup.

Run rate improves on the seasonal baseline

1.50%
run-rate MAPE

2.20%
seasonal-profile MAPE

5 / 8
quarters beating seasonal



Remaining-output MAPE: 3.81% vs 5.54%. Small sample, publication proxies and correlated cutoffs limit confidence.

Tax effects and reinvestment complicate the buy case

Evidence	Period	Result
Owners’ profit growth	FY26	+15.5% YoY
Revenue from operations	FY26	−0.4% YoY
Owners’ profit	Q1 FY27	₹6,721.05 crore, +11.8% YoY
Borrowings, excluding leases	30 Jun 2026	₹2,72,081.67 crore
Company debt-service coverage	FY26	1.23×

A deferred-tax credit interacts with regulatory movements. Verified normalized consolidated EPS remains an open research question.

Operating cash does not cover capex plus interest

FY26 selected cash bridge, ₹ crore



₹6,852 crore after capex becomes −₹10,686 crore after cash interest.

Interest sits in financing. The selected residual after dividends excludes other flows and is not a complete cash reconciliation.

The entry-price screen exposes the assumptions

Dated NSE close

₹303.75

10.61× reported trailing P/E

Example entry threshold

₹248.22

Wait under these assumptions

Example assumptions

15% earnings haircut, 12× assumed fair P/E, 15% margin of safety.

What changes the result

Editable inputs change the screen. Recurring earnings and a funding plan still need to support the thesis.

TTM EPS is reported, not normalized. The screen evaluates assumptions and does not predict investment returns.

Prismy answers from the site’s evidence

A bounded answer surface

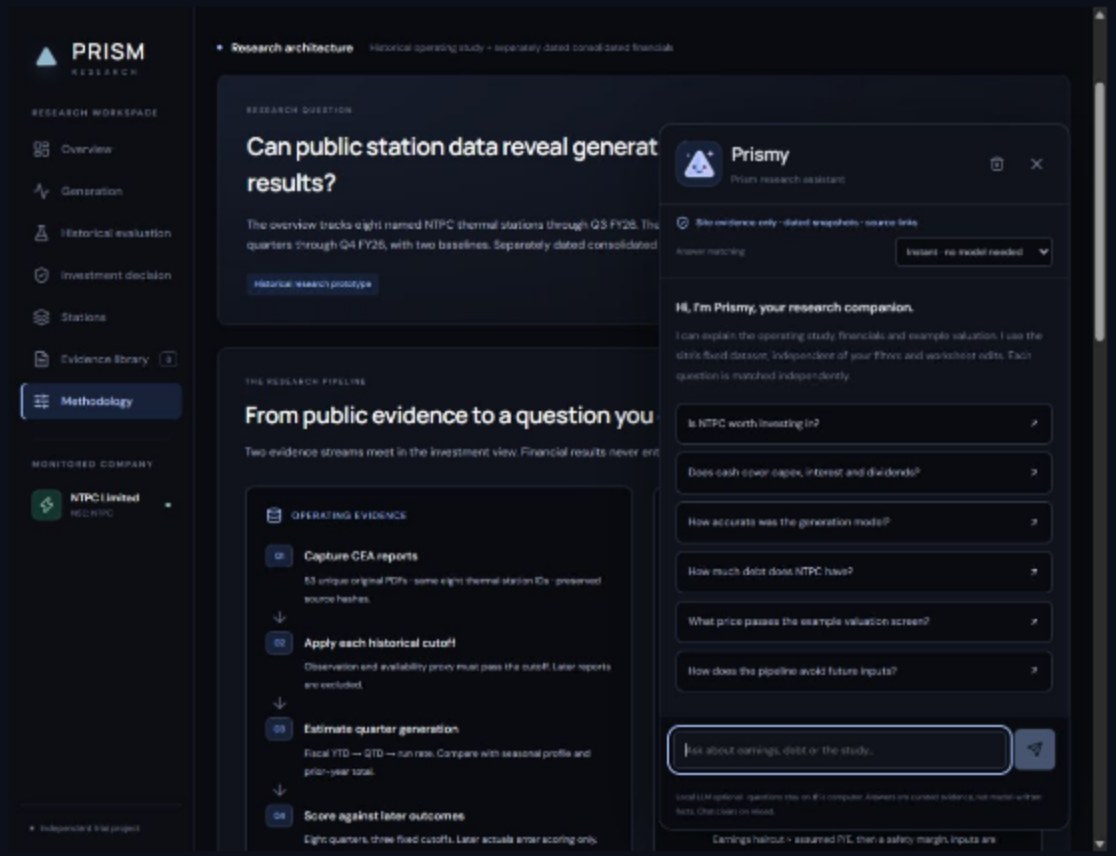
Six preset questions and typed queries over 14 canonical evidence cards.

Sources accompany answers

Unknown questions receive an explicit evidence-gap response.

Optional model routing

Local Ollama selects evidence IDs. Static hosting uses instant matching.



The most useful findings changed the design

A stronger baseline changes the claim

The seasonal profile wins three of eight day-60 quarters. I retained every quarter and all cutoff comparisons.

Known output makes headline error look better

I added remaining-output error so the reader can assess the genuinely unobserved part of the quarter.

Cash classification changes the interpretation

I subtracted financing interest explicitly before discussing cash available after reinvestment.

Restricted generation improves factual control

Prismy renders source-backed cards. The LLM can match a question but cannot write new factual claims.

The next version would strengthen provenance and economics

First priority: verify historical availability

Capture first-public timestamps and revision vintages, then reserve prospective quarters before testing.

Second: expand coverage consistently

Map legal entities, technologies and commissioning events across the full fleet.

Third: model recurring earnings and funding

Separate tariff economics, normalize interacting adjustments, test project returns and financing needs.

Fourth: broaden assistant evaluation

Use a larger paraphrase and unsupported-question set. Add a secured hosted model only if it is useful.

A reviewer can reproduce the result

Live dashboard

https://ronnismad.github.io/prism-ntpc-research-dashboard/

Source repository

https://github.com/ronnismad/prism-ntpc-research-dashboard

Suggested five-minute review

Historical evaluation, Investment decision, Methodology, then a Prismy question and a source link.

Reproducible handoff

28 tests, source hashes, full-precision CSVs, README and local setup.

Prepared by Sutanu Chatterjee for the PRISM trial task. Data and market quote are dated snapshots.